Table II: SysJava vs SWAT4J on the primary JDK 8 container images (Debian 10, Temurin 8u504).

| | Zookeeper 3.4.14 JDK 8u504 Debian 10 | Tomcat 7.0 JDK 8u504 Debian 10 | Cassandra 3.0.29 JDK 8u504 Debian 10 | Jetty 9.4.51 JDK 8u504 Debian 10 | Storm 2.4.0 JDK 8u504 Debian 10 | OrientDB 3.2.19 JDK 8u504 Debian 10 | Elasticsearch 7.5.2 JDK 8u504 Debian 10 | Solr 8.7.0 JDK 8u504 Debian 10 | Groovy 4.0.0 JDK 8u504 Debian 10 | JRuby 9.4.2.0 JDK 8u504 Debian 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Native Method Resolution | | | | | | | | | | |
| JNI Total Natives | 495 | 816 | 975 | 353 | 1884 | 700 | 626 | 980 | 588 | 598 |
| SysJava Resolved | 484 | 792 | 903 | 352 | 1835 | 684 | 609 | 967 | 541 | 587 |
| JNI Static (Java_*) | 317 | 564 | 596 | 185 | 1347 | 498 | 427 | 645 | 360 | 416 |
| Partial (_1/$/__) | 4 | 59 | 31 | 2 | 65 | 18 | 14 | 8 | 7 | 6 |
| JNI Dynamic (RegisterNatives) | 119 | 122 | 226 | 119 | 366 | 123 | 121 | 261 | 122 | 120 |
| JVM | 44 | 47 | 45 | 46 | 47 | 45 | 47 | 47 | 52 | 45 |
| JFR/Netty | 0 | 0 | 5 | 0 | 10 | 0 | 0 | 6 | 0 | 0 |
| Platform / Intrinsic | 10 | 21 | 71 | 0 | 46 | 8 | 16 | 10 | 39 | 3 |
| Not Found | 1 | 3 | 1 | 1 | 3 | 8 | 1 | 3 | 8 | 8 |
| SWAT4J Resolved | 198 | 500 | 534 | 162 | 1257 | 377 | 287 | 277 | 271 | 366 |
| SWAT4J JNI Static (Java_*) | 161 | 459 | 388 | 122 | 1090 | 338 | 247 | 237 | 228 | 327 |
| JNI Partial/Heuristic | 0 | 0 | 107 | 0 | 127 | 0 | 0 | 0 | 0 | 0 |
| JVM Source Code | 37 | 41 | 39 | 40 | 40 | 39 | 40 | 40 | 43 | 39 |
| Not Found | 290 | 308 | 434 | 184 | 619 | 315 | 331 | 693 | 308 | 225 |
| Syscall Allowlist (SysJava only) | | | | | | | | | | |
| Java Lib Syscalls | 108 | 124 | 124 | 92 | 125 | 111 | 113 | 127 | 109 | 95 |
| Container Binary Syscalls | 111 | 92 | 140 | 63 | 101 | 83 | 114 | 111 | 105 | 84 |
| Go Whitelist (Δ) | 36 | 35 | 24 | 51 | 26 | 40 | 31 | 28 | 37 | 43 |
| Unmapped Lib Syscalls (Δ) | 18 | 0 | 41 | 10 | 2 | 10 | 11 | 20 | 19 | 10 |
| Total Syscalls | 190 | 170 | 230 | 157 | 172 | 173 | 183 | 199 | 189 | 167 |
| SWAT4J Syscall Comparison | | | | | | | | | | |
| SWAT4J Java Lib Syscalls | 104 | 105 | 116 | 104 | 106 | 105 | 105 | 103 | 108 | 104 |
| SWAT4J Dynamic Syscalls | 128 | 127 | 141 | 128 | 137 | 129 | 137 | 136 | 130 | 128 |
| SWAT4J Total (Static ∪ Dynamic) | 171 | 170 | 179 | 171 | 177 | 169 | 174 | 173 | 175 | 170 |
| SysJava ∩ SWAT4J | 136 | 134 | 159 | 125 | 138 | 133 | 139 | 144 | 140 | 133 |
| SysJava only | 54 | 36 | 71 | 32 | 34 | 40 | 44 | 55 | 49 | 34 |
| SWAT4J only | 35 | 36 | 20 | 46 | 39 | 36 | 35 | 29 | 35 | 37 |
| Dynamic Capture Artefacts | | | | | | | | | | |
| Libraries Loaded | 20 | 24 | 119 | 15 | 29 | 23 | 35 | 28 | 19 | 19 |
| Binaries Loaded | 11 | 6 | 23 | 2 | 10 | 6 | 9 | 14 | 12 | 7 |
| JARs Loaded | 10 | 41 | 51 | 16 | 88 | 76 | 182 | 144 | 69 | 9 |
| Directly Mapped Libraries Analysed | 11 | 15 | 19 | 8 | 15 | 18 | 15 | 18 | 15 | 7 |

Note: SysJava Resolved + Platform / Intrinsic + Not Found = JNI Total Natives; SWAT4J Resolved + Not Found + its intrinsic natives = JNI Total Natives (SWAT4J's mapping run on the same native methods and image libraries). Java Lib ∪ Container Binary + Go Whitelist (Δ, runc entries not already counted) + Unmapped Lib (Δ, syscalls of loaded libraries without a Java binding, not already counted) = Total Syscalls. Native methods are found on an RTA call graph (virtual and interface calls dispatched to the overrides of every class instantiated in reachable code). Natives in libraries the image does not ship are mapped to the released library (Solr: Hadoop 3.2.0 libhadoop, Hadoop 3.3.6 libhadoop for the 14 erasure-coding natives, netty epoll 4.1.50, Lucene NativePosixUtil; OrientDB: jffi 1.3.11, NativePosixUtil; Elasticsearch: NativePosixUtil; Groovy: the jansi library bundled in jline 2.14.6; Tomcat: jsvc built from commons-daemon 1.2.4); each library is analysed once from all its mapped start functions. The remaining Not Found natives have no implementation in any library (e.g. jffi Foreign natives never implemented upstream, JFR setCompressedIntegers absent from the 8u504 libjvm). Container binaries are analysed from _start and main; runc is covered by the Go whitelist only. Storm: sync in its generated profile comes only from FFM hits the detector inferred from Clojure class names

(JDK 8 has no FFM API) and is not counted; Storm runs under the 172-syscall profile. Elasticsearch: SysPart cannot analyse the stripped ML controller; its syscalls are recovered from the functions it imports. SWAT4J Java Lib = its static allowlist (JTracer); SWAT4J Dynamic = strace of the container start-up. All ten containers run under their generated profiles.